

TECHNICAL UNIVERSITY OF CRETE

School of Electrical & Computer Engineering

Investigating the Applicability of Federated Dynamic Averaging in Federated Low-Rank Adaptation Architectures

Diploma Thesis Presentation

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Introduction : Federated Fine-Tuning of Foundation Models

- **Federated Learning (FL)**: Privacy-preserving collaborative model optimization across decentralized edge devices or institutional silos without centralizing raw data.
- **Foundation Model Scaling**: Adaptation of large pre-trained transformer models (e.g., RoBERTa 125M/355M, Llama-2 7B) to downstream domains is vital across healthcare, finance, and legal domains.
- **The Core Constraint**: Transmitting billions of parameters over bandwidth-limited edge networks makes standard centralized or full-parameter federated learning computationally and communicatively intractable.

Central Research Question

How can we jointly optimize both the **spatial parameter volume** and the **temporal synchronization frequency** in federated learning without sacrificing global aggregation fidelity?

Introduction : The Dual Communication Bottleneck

1. Spatial Bottleneck

- **Physical Volume**: Data transferred per synchronization round.
- Full parameter transfers require gigabytes per client per round (e.g., 28 GB for Llama-2-7B in FP32).
- Addressed via alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025).

2. Temporal Bottleneck

- **Synchronization Cadence**: Frequency of communication rounds.
- Static schedules ($\tau = E$) ignore local optimization dynamics and statistical data drift.
- Addressed in full models via dynamic averaging (**FDA-Opt**, Theologitis et al., 2025).

True communication efficiency requires co-optimizing both dimensions simultaneously:
***FDA-Opt** (Temporal) + **RoLoRA** (Spatial).*

Introduction : Core Objectives & Scientific Contributions

- **Conceptual Integration**: First formulation uniting temporal dynamic averaging (**FDA-Opt**, Theologitis et al., 2025) with alternating parameter-efficient adaptation (**RoLoRA**, Chen et al., NeurIPS 2025).
- **FDA-RoLoRA Framework**: Embeds variance-driven dynamic synchronization into RoLoRA's alternating optimization scheme with independent per-matrix variance thresholds (θ_A, θ_B).
- **Direct Subspace Variance Tracking**: Formulates variance tracking operating strictly within low-rank coordinate tensors, avoiding full-parameter expansion overhead.
- **Decoupled Subspace Threshold Evolution**: Dynamically updates synchronization thresholds for each matrix track independently via midpoint scaling, avoiding cross-phase geometric interference.
- **Software Architecture & Foundation Model Testbed**: Extended the open-source **FDA-Opt** codebase to build an end-to-end framework on PyTorch, Transformers, and PEFT, benchmarked on foundation models (`roberta-base`) across GLUE tasks on an enterprise GPU cluster.

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Background : Distributed Learning, FedAvg & Client Drift

- **Federated Averaging (FedAvg)**:

$$w_{t+1} = w_t + \frac{1}{|S_t|} \sum_{k \in S_t} \Delta_t^{(k)}$$

  Clients execute τ local SGD steps on private data $\mathcal{D}_k$.
- **Statistical Heterogeneity (Non-IID)**: Client distributions differ in label, quantity, or features. Modeled via **Dirichlet distribution** $p_k \sim \text{Dir}(\alpha)$.
- **Client Drift**: Divergent local minima pull updates away from the true global optimum.

Dirichlet Heterogeneity (α)

- $\alpha \to 0$: Extreme label skew (clients hold single-class data).
- $\alpha \to \infty$: Uniform IID distribution.
- Key benchmark lever for evaluating drift robustness.

Background : Low-Rank Adaptation (LoRA) & Factor Asymmetry

- Pre-trained weights $W_0 \in \mathbb{R}^{d \times k}$ are kept **frozen**.
- Updates factorized into low-rank matrices:

$$\Delta W = AB, \quad A \in \mathbb{R}^{d \times r},\ B \in \mathbb{R}^{r \times k}$$

  where rank $r \ll \min(d, k)$.
- Forward pass: $h = W_0 x + ABx$.
- Trainable parameters reduced by $> 99\%$.
- **Factor Asymmetry**:
  - A: Down-projection (compression), Gaussian init.
  - B: Up-projection (expansion), zero init.
  - Exhibits distinct gradient flows and variance dynamics.

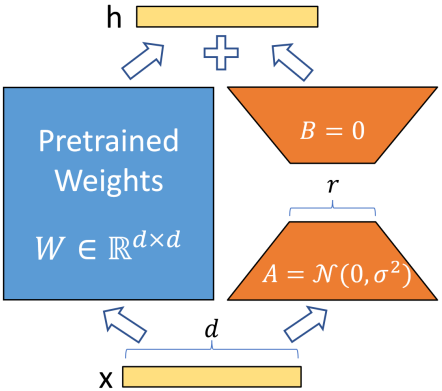


Figure: LoRA decomposition with frozen weights W_0.

Background : The LoRA Aggregation Inconsistency Problem

Mathematical Non-Commutativity (Bilinear Distortion)

Independent server-side averaging of client low-rank factors fails to reconstruct the composite weight update:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i B_i) \neq \left(\frac{1}{N}\sum_{i=1}^{N} A_i\right)\left(\frac{1}{N}\sum_{i=1}^{N} B_i\right)$$

- Direct FedAvg on (A_i, B_i) induces cross-client geometric interference.
- Destabilizes convergence under non-IID data distributions.
- **Prior Work Limitations**:
  - *Full-matrix expansion* ($\Delta W = AB$): Transmits full parameter matrices $\rightarrow$ destroys spatial savings.
  - *FFA-LoRA*: Permanently freezes down-projection A $\rightarrow$ sacrifices adapter expressivity.
  - *FLoRA*: Server-side SVD recalculation $\rightarrow$ high server compute overhead.

Alternating Coordinate Optimization

RoLoRA resolves the LoRA bilinear aggregation inconsistency by alternating the active trainable matrix while freezing the other globally.

- **Odd Rounds (B-Track)**: Freeze A^t; train B:

$$\frac{1}{N}\sum_{i=1}^{N}(A^t B_i^{t+1}) = A^t \bar{B}^{t+1} = A^t B^{t+1}$$

- **Even Rounds (A-Track)**: Freeze B^{t+1}; train A:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i^{t+1} B^{t+1}) = \bar{A}^{t+1} B^{t+1} = A^{t+1} B^{t+1}$$

- **Limitation**: Enforces a rigid, static local epoch schedule (E), leaving temporal synchronization unoptimized.

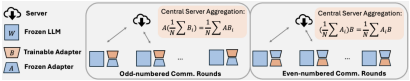


Figure: RoLoRA alternating framework.

Background : FDA-Opt & Geometric Monitoring

Theologitis, Samoladas & Deligiannakis, 2025 – Dynamic Variance Halting

- **Geometric Monitoring Root**: Decomposes global threshold functions over distributed streams into local **safe zones** (Sharfman et al., 2006).
- **FDA-Opt Formulation** (Theologitis et al., 2025): Tracks client update dispersion via parameter variance:

$$\mathrm{Var}(W_{t,i}) = \frac{1}{|S_t|} \sum_{k \in S_t} \|\Delta_{t,i}^{(k)}\|_2^2 - \|g_{t,i}\|_2^2$$

  Low variance = aligned progress; high variance = client drift.
- **Event-Driven Halting**: Synchronize when $\mathrm{Var}(W) > \theta_t$.
- **Midpoint Threshold Scaling**: $\theta_{t+1} = \mathrm{ar}_t \cdot \frac{\tau}{2s_t}$.

Research Gap in FDA-Opt

- FDA-Opt operated strictly on **full model parameters** (approximated via AMS sketches).
- Spatial compression (LoRA) and temporal adaptivity (FDA) had never been synthesized.
- Low-rank adapters exhibit high variance volatility requiring decoupled mechanics.

Background : Comparison of Foundational Precursors

Synthesizing RoLoRA (Spatial) and FDA-Opt (Temporal)

Dimension	**RoLoRA** (NeurIPS '25)	**FDA-Opt** (2025)	**FDA-RoLoRA** (**This Thesis**)
Optimization Focus	Spatial parameter volume	Temporal synchronization	**Joint Spatial + Temporal**
Parameter Scope	Low-rank adapters (A, B)	Full model weights	**Low-rank adapters (A, B)**
Aggregation	Alternating coordinates	Standard FedAvg / FedOpt	**Alternating coordinates**
Local Scheduling	Static fixed (E epochs)	Dynamic (Model variance)	**Dynamic (Subspace variance)**
Variance Tracking	None	Full-weight (AMS sketch)	**Exact Subspace** (No sketch)
Threshold Logic	None	Single scalar threshold	**Decoupled Dual-Track** (θ_A, θ_B)

Complementary Synergy

FDA-RoLoRA solves the core limitations of both parent papers: RoLoRA provides the strict aggregation linearity needed for stable variance tracking, while FDA-Opt provides the event-driven cadence needed to eliminate static schedule waste.

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Methodology : Dynamic Averaging in Federated LoRA

Taxonomy of Integration Avenues

Different PEFT architectures have distinct mathematical structures and optimization geometries; dynamic averaging cannot use a single one-size-fits-all rule.

- **FDA + FedAvg-LoRA**: Dynamic averaging with simultaneous (A, B) training.
  - *Drawback*: Inherits bilinear aggregation error; non-linear distortion destabilizes variance monitoring.
- **FDA + FFA-LoRA**: Dynamic averaging with permanently frozen A *[Future Work]*.
- **FDA + SLoRA**: Dynamic thresholding over sparse adapter coordinates *[Future Work]*.
- **FDA-RoLoRA**: **This Thesis**
  - Synthesizes **RoLoRA** (Chen et al., 2025) and **FDA-Opt** (Theologitis et al., 2025).
  - Alternating coordinates provide strict linearity $\rightarrow$ perfect substrate for variance tracking.
  - Maintains independent persistent variance thresholds (θ_A, θ_B) for each subspace.

Methodology : FDA-RoLoRA Architecture

Decoupled Subspace Variance Mechanics

RoLoRA isolates the optimization trajectories of down-projection (A) and up-projection (B). FDA-RoLoRA leverages this to isolate variance tracking and threshold evolution.

- **Active Subspace Mapping**: Server maps active threshold based on designated matrix $\mathcal{M}^{(r)} \in \{A, B\}$:

$$\theta_{\text{current}}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

- **Direct Subspace Variance**: Evaluated exclusively on low-rank coordinate tensors:

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)}\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

- Eliminates memory/compute overhead of mapping drifts back to the full weight space.

Methodology : Independent Halting & Local Training Bounds

Local Epoch Bounds

Client training is bounded between $[e_{\min}, e_{\max}]$ to prevent extreme stagnation or runaway drift:

$$\text{Halt if } \mathrm{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}} \ \vee \ e \geq e_{\max}$$

Evaluated at epoch boundaries $e \geq e_{\min}$.

Target Threshold Proposal

Targets midpoint training progress $(e_{\max}/2)$:

$$\theta^{(r)}_{\text{target}} = \left(\frac{e_{\max}/2}{e^*} \right) \mathrm{Var}_{\text{subspace}}(\Delta^{(e^*)})$$

where e^* is the realized epoch count at termination.

Prevention of Cross-Matrix Contamination

The inactive matrix threshold remains strictly frozen during the alternate round, preserving optimization state and respecting factor asymmetry.

Methodology : Threshold Evolution Mechanics & Controller Design

Primary Operational Controller: Standard Midpoint Scaling

Direct assignment without memory lag: $\theta_{\mathcal{M}^{(r)}}^{(\text{next})} = \theta_{\text{target}}^{(r)}$.

- **Parameter-Free**: No sensitivity to tuning hyperparameters.
- **Adaptive Response**: Responds instantly to changes in gradient drift variance.

Exploratory Temporal Smoothing Variants (Prototyped)

- **Simple Moving Average (SMA)**: Mean over past K active rounds: $\frac{1}{K}\sum_{k=0}^{K-1}\theta_{\text{target}}^{(r-k)}$.
- **Exponential Moving Average (EMA)**: Historical weighting: $\rho\theta_{\text{current}} + (1-\rho)\theta_{\text{target}}$.
- **Finding**: Prototyping revealed smoothing added threshold inertia/lag under distribution shifts without tangible accuracy benefit; hence retained as future work.

Methodology : Server Adaptive Optimization & Protocol Flow

Server FedAdamW Optimizer

- Aggregated pseudo-gradient:

$$\nabla_{\text{pseudo}}^{(r)} = -\frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$$

- First & second moment updates:

$$m^{(r)} = \beta_1 m^{(r-1)} + (1-\beta_1)\nabla_{\text{pseudo}}^{(r)}$$
$$v^{(r)} = \beta_2 v^{(r-1)} + (1-\beta_2)(\nabla_{\text{pseudo}}^{(r)})^2$$

- Decoupled weight decay:

$$W_{\text{act}}^{(r+1)} = W_{\text{act}}^{(r)} - \eta_s \left(\frac{\hat{m}^{(r)}}{\sqrt{\hat{v}^{(r)}} + \epsilon} + \lambda W_{\text{act}}^{(r)} \right)$$

Protocol Execution Lifecycle

1. **Toggle**: Server selects active track A or B.
2. **Broadcast**: Sends $W_{\text{act}}^{(r)}$ and $\theta_{\text{current}}^{(r)}$ to S_r.
3. **Train**: Clients run local SGD steps.
4. **Check**: Evaluate subspace variance at $e \geq e_{\min}$.
5. **Halt**: Return $\Delta_i^{(e^*)}$ when $\text{Var} > \theta$ or $e = e_{\max}$.
6. **Aggregate**: Server forms pseudo-gradient.
7. **Evolve**: Update active threshold $\theta_{\mathcal{M}^{(r)}}^{(\text{next})}$.
8. **Step**: Server AdamW updates active weights.

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Implementation : Modular 3-Tier Software Architecture

1. Server Orchestrator

- Global A, B matrices.
- Tracks (θ_A, θ_B) state.
- Coordinates RoLoRA toggle.
- FedAdamW server optimizer.

2. Client Nodes

- Private data partitions.
- Local SGD training loops.
- Subspace drift computation.
- Variance checks at $e \geq e_{\min}$.

3. Comm. Interface

- Low-rank serialization.
- PyTorch, HF & PEFT.
- Lightweight FDA-Opt core.

Key Engineering Decisions

- **FDA-Opt Base vs. FederatedScope-LLM**: The RoLoRA paper (Chen et al., NeurIPS 2025) used FederatedScope-LLM without a public release. We adopted and extended the open-source **FDA-Opt** repository (Theologitis et al., 2025), integrating HuggingFace PEFT and RoLoRA coordinate alternation into its modular PyTorch core.
- **Exact Variance over AMS Sketch**: Full-parameter FDA-Opt required randomized AMS sketching to bound memory. In low-rank coordinates ($r = 8$), parameters are so compact that exact subspace variance is computed with negligible overhead, eliminating sketching approximations.

Implementation : Foundation Model Setup & Data Partitioning

Foundation Model Backbone

- **Model**: `roberta-base` (125M params, 12 layers, 768 hidden; selected for multi-seed sweeps under hardware budget).
- **Adapter Placement**: Low-rank adapters injected into Query (W_q) and Value (W_v) projections across top encoder layers.
- **Adapter Configuration**: Rank $r = 4$, scaling factor $\alpha_{\text{lora}} = 8$ (0.47% trainable parameter share).
- **Spatial Reduction**: Backbone frozen; communicating only compact A, B factors cuts parameter transfer by $> 99\%$.

Resource Constraints & Simulation Setup

- **Dirichlet Partitioning**: Client data sampled via $p_k \sim \text{Dir}(\alpha)$ ($\alpha \in \{0.5, 1000.0\}$).
- **25k Dataset Cap**: Datasets capped at $\leq 25{,}000$ samples (MAX_SAMPLES $= 25{,}000$) due to compute limits.
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Simulation Early Stopping**: Accuracy-gated early stopping to prevent redundant GPU-hours.
- **Benchmark Tasks**: GLUE suite (SST-2, QQP, QNLI, MNLI).

Implementation : Training Pipeline & Dedicated Testbed Deployment

Training Pipeline Mechanics

- **Alternation**: Track B on even rounds ($t = 0, 2, \dots$), Track A on odd rounds ($t = 1, 3, \dots$).
- **Exact Subspace Variance**: Evaluated centrally at $e \geq e_{\text{min}}$ without high-dimensional projection.
- **Optimization**: Client SGD ($\eta_c = 0.1$), server SGD step ($\eta_s = 1.0$) replicating FedAvg over active coordinate.
- **Orchestration**: Slurm cluster jobs supported; multi-process client worker orchestration.

Polytechnix Server Testbed

- **Server**: SoftNet Lab Polytechnix GPU server.
- **GPUs**: 2× NVIDIA Ampere A10 (24 GB VRAM each).
- **CPUs**: 2× Intel Xeon Silver 4310 (24c/48t, 2.10 GHz).
- **RAM**: 256 GB DDR4 system memory.
- **Comparability**: All 173 runs evaluated exclusively on this uniform platform to ensure perfect comparability.

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Evaluation : Experimental Setup & Benchmark Protocol

Model & Low-Rank Parameterization

- **Backbone**: `roberta-base` (125M params, 12 layers).
- **LoRA Setup**: Rank $r = 4$, $\alpha = 8$ on attention W_q, W_v (0.47% trainable parameters).
- **Uplink Payload**: 240 KB/client/round (sending only active factor A or B).
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Optimization**: Mini-batch SGD ($\eta_c = 0.1$, $\eta_s = 1.0$).

Benchmark Matrix & Regimes

- **Tasks**: GLUE suite (MNLI, QNLI, QQP, SST-2) capped at 25,000 training samples.
- **Cohorts**: $C \in \{20, 50\}$ clients (full participation).
- **Regimes**: Homogeneous ($\alpha = 1000.0$) vs. Dirichlet Label Skew ($\alpha = 0.5$).
- **Baselines**: Fixed RoLoRA $E \in \{5, 10, 15, 20\}$.
- **FDA-RoLoRA**: $[e_{\min}, e_{\max}] = [1, 20]$ (and $[1, 40]$), scaling factor $k \in \{0.5, 1.0\}$.
- **Scale**: **173 completed runs** across 86 configs.

Architectural Capacity Context

Validation accuracies reflect the 125M-parameter `roberta-base` capacity. We perform an exact head-to-head comparison isolating the algorithmic contribution of dynamic variance tracking.

Evaluation : Executive Performance Scorecard

Comprehensive Synthesis Across GLUE Benchmarks on `roberta-base`

Task & Cohort	Regime	Top FDA-RoLoRA	FDA Peak Acc. (%)	Top RoLoRA Baseline	RoLoRA Peak (%)	Δ Acc.
MNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	$\mathbf{76.86 \pm 0.07}$	RoLoRA ($E = 10$)	76.60 ± 0.60	**+0.26%**
	Non-IID ($\alpha = 0.5$)	FDA ($k = 1.0$)	$\mathbf{74.47 \pm 0.08}$	RoLoRA ($E = 10$)	74.24 ± 0.09	**+0.23%**
MNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	$\mathbf{69.85 \pm 2.38}$	RoLoRA ($E = 10$)	65.85 ± 2.02	**+4.00%**
	Non-IID ($\alpha = 0.5$)	FDA ($k = 0.5$)	55.56 ± 1.17	RoLoRA ($E = 10$)	$\mathbf{56.55 \pm 4.90}$	-0.99%
QNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	$\mathbf{83.14 \pm 0.24}$	RoLoRA ($E = 10$)	81.83 ± 0.65	**+1.31%**
	Non-IID ($\alpha = 0.5$)	FDA ($k = 1.0$)	$\mathbf{84.22 \pm 0.29}$	RoLoRA ($E = 10$)	83.50 ± 1.18	**+0.72%**
QNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	82.77 ± 1.51	RoLoRA ($E = 20$)	$\mathbf{88.14 \pm 0.00}$	-5.37%
	Non-IID ($\alpha = 0.5$)	FDA ($k = 0.5$)	$\mathbf{75.69 \pm 1.17}$	RoLoRA ($E = 10$)	74.61 ± 7.18	**+1.08%**
QQP ($C = 20$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	$\mathbf{78.57 \pm 2.99}$	RoLoRA ($E = 10$)	75.47 ± 0.84	**+3.10%**
	Non-IID ($\alpha = 0.5$)	FDA ($k = 1.0$)	$\mathbf{77.88 \pm 0.46}$	RoLoRA ($E = 10$)	77.69 ± 1.54	**+0.19%**
SST-2 ($C = 50$)	IID ($\alpha = 1000$)	FDA ($k = 1.0$)	89.68 ± 0.80	RoLoRA ($E = 20$)	$\mathbf{92.01 \pm 0.46}$	-2.33%
	Non-IID ($\alpha = 0.5$)	FDA ($k = 1.0$)	$\mathbf{85.95 \pm 3.96}$	RoLoRA ($E = 10$)	59.00 ± 7.86	**+26.95%**

Key Empirical Breakthroughs

Anti-Collapse on SST-2 Non-IID: RoLoRA crashes to 59.00%; FDA secures **85.95%** (+**26.95%** recovery).
Consensus Acceleration on MNLI: Breaks static plateau at 65.85% to **69.85%** (+**4.00%** outperformance).
Communication Reductions: Converges with up to **56.6% lower network bandwidth** (QNLI Non-IID $C = 50$).

Evaluation : Homogeneous Consensus (IID, $\alpha = 1000.0$)

The Compute Accelerator Mechanism in Shared-Gradient Environments

Optimization Dynamics

- **Aligned Gradients**: Client updates point along near-consensus trajectories toward global optimum.
- **Suppressed Variance**: Coordinate variance remains low ($\bar{\nu} \leq 1.0$).
- **Safe Epoch Expansion**: Controller elevates θ, expanding local training to $\bar{e} \approx 11$–13 epochs per round.

Empirical Highlights

- **QQP ($C = 50$)**: Early stops in **14.0 rounds** (164.1 MB) vs. 30.0 rounds (351.6 MB) $\rightarrow$ **53.3% bandwidth savings**.
- **MNLI ($C = 50$)**: Advances to **69.85**% vs. RoLoRA's 65.85% plateau (+**4.00**%).
- **QNLI ($C = 20$)**: Peak accuracy **83.14**% vs. 81.83%.

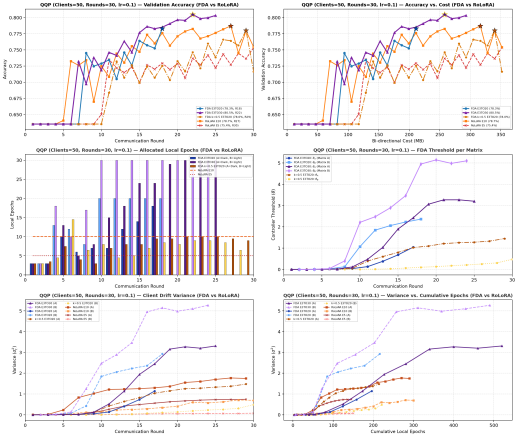


Figure: Multi-panel telemetry on QQP ($C = 50$, IID): (a) Acc/Rounds; (b) Acc/MB; (c) Epochs; (d) θ; (e) $\bar{\nu}$; (f) Var/Epochs.

Evaluation : Statistical Heterogeneity (Non-IID, $\alpha = 0.5$)

The Safety Brake Mechanism & Anti-Collapse Protection Under Label Skew

The "Safety Brake" Mechanism

- Dirichlet skew ($\alpha = 0.5$) causes drift variance to surge **4.0**×.
- Static schedules force deep epochs → post-peak collapse.
- FDA controller senses spikes and **throttles local epochs**, preserving decision boundaries.

Empirical Highlights

- **SST-2 (**$C = 50$**)**: RoLoRA crashes to 59.00%; FDA reaches **85.95**% (+**26.95**% recovery).
- **QNLI (**$C = 20$**)**: Suite-leading **84.22**% peak accuracy.
- **QNLI (**$C = 50$**)**: Early stops in 13.0 rd (152.3 MB) vs. 30.0 rd → **56.6**% bandwidth savings.

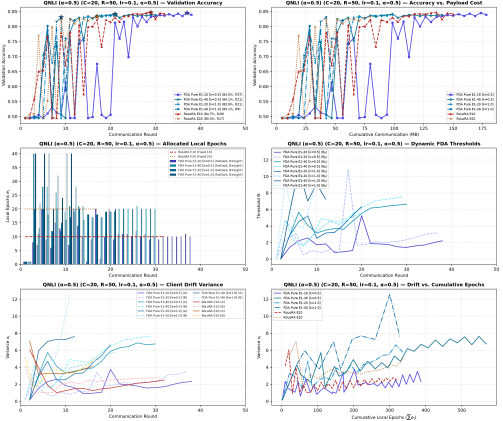


Figure: Multi-panel telemetry on QNLI under Dirichlet skew ($\alpha = 0.5$, $C = 20$): closed-loop tracking securing 84.22% accuracy.

Evaluation : Factor Asymmetry & Decoupled Tracking

Down-Projection (A) vs. Up-Projection (B) Optimization Geometry

Mathematical Factor Asymmetry

- **Down-Projection** $A \in \mathbb{R}^{d \times r}$: Compresses input features into low-rank bottleneck; Gaussian initialization.
- **Up-Projection** $B \in \mathbb{R}^{r \times d}$: Expands bottleneck representations back to hidden dimension; zero initialization.
- Exhibit distinct gradient magnitudes and variance volatility.

Decoupled Threshold Isolation

Maintaining independent thresholds (θ_A, θ_B) ensures that the inactive track remains frozen, eliminating cross-phase geometric interference.

Task & Cohort	Config	Mean $\bar{e}$	$\bar{e}_A$	$\bar{e}_B$
MNLI ($C = 20$, Non-IID)	$k = 1.0$	17.49	17.19	**17.76**
MNLI ($C = 50$, Non-IID)	$k = 0.5$	10.13	9.47	**10.80**
QNLI ($C = 20$, Non-IID)	$k = 1.0$	16.73	16.63	**16.81**
QNLI ($C = 50$, Non-IID)	$k = 0.5$	10.48	9.83	**11.13**
QQP ($C = 50$, Non-IID)	$k = 1.0$	14.26	13.83	**14.63**

Key Finding

Up-projection factor B consistently absorbs deeper local epoch allocations ($\bar{e}_B > \bar{e}_A$) across benchmarks, confirming that low-rank factors require decoupled dynamic regulation.

Evaluation : Communication Efficiency: Milestone Gates

Theologitis-Style Analysis: Rounds and Total Payload to Target Accuracies

Task ($C = 20$)	Regime	Methodology	Milestone Gate 1	Milestone Gate 2	Milestone Gate 3	Total MB
SST-2	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 23 rd (107.8 MB)	R@85%: 24 rd (112.5 MB)	R@88%: 24 rd (112.5 MB)	117.2 MB
		FDA-RoLoRA ($k = 1.0$)	**R@80%: 8 rd (37.5 MB)**	**R@85%: 8 rd (37.5 MB)**	**R@88%: 9 rd (42.2 MB)**	**65.6** MB
MNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@65%: 28 rd (131.2 MB)	R@72%: 43 rd (201.6 MB)	R@74%: 50 rd (234.4 MB)	234.4 MB
		FDA-RoLoRA ($k = 1.0$)	**R@65%: 19 rd (89.1 MB)**	**R@72%: 25 rd (117.2 MB)**	**R@74%: 36 rd (168.8 MB)**	**178.1** MB
QNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 17 rd (79.7 MB)	R@83%: 25 rd (117.2 MB)	R@84%: — (Unattained)	154.7 MB
		FDA-RoLoRA ($k = 1.0$)	**R@80%: 15 rd (70.3 MB)**	**R@83%: 21 rd (98.4 MB)**	**R@84%: 25 rd (117.2 MB)**	**124.2** MB

2.87× Convergence Speedup (SST-2)

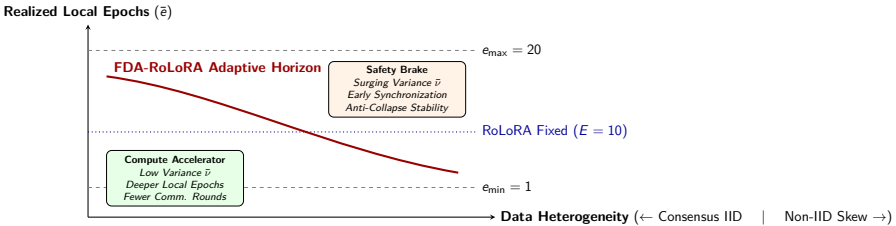
Crosses 80% in **8 rounds** (37.5 MB) vs. 23 rounds (107.8 MB) → **65.2% bandwidth savings**.

Higher Generalization Ceilings (QNLI)

Reaches 84% gate in **25 rounds** (117.2 MB), whereas static RoLoRA fails to ever cross 84%.

Evaluation : Cross-Regime Synthesis: The Dual Frontier

Autonomous Operating Modes Governed by Coordinate Variance



Consensus Accelerator: Low drift → scales $\theta \to \bar{e} \to e_{max}$, slashing round latency.

Anti-Drift Safety Brake: High drift → throttles $\bar{e} \to e_{min}$, preventing catastrophic collapse.

Evaluation : Practical Deployment Guidelines

Four Core Architectural Principles for Real-World Federated Systems

1. Guarantee Base Progress ($e_{\min} = 1$)

Setting $e_{\min} \geq 1$ ensures clients complete meaningful SGD steps before variance checks can trigger, preventing round stagnation.

2. Horizon Selection (k)

- $k = 1.0$ **(Aggressive)**: Best for homogeneous / cross-silo setups to maximize compute and cut rounds.
- $k = 0.5$ **(Midpoint)**: Best for acute non-IID edge networks to strictly contain client drift.

3. Decouple Factor Thresholds

Always maintain independent (θ_A, θ_B) states. Freezing the inactive threshold respects low-rank factor asymmetry.

4. Calibrate Early Stopping Under Skew

In non-IID regimes, configure wider smoothing windows ($W \geq 5$) and a round floor ($R_{\min} \geq 0.6 \times R_{\max}$) to tolerate alternating validation oscillations.

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Conclusion : Synthesis of Scientific Contributions

- **Bridging RoLoRA & FDA-Opt**: Unified alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025) with dynamic variance-driven averaging (**FDA-Opt**, Theologitis et al., 2025) into a coherent dual-axis federated framework.
- **Mathematical Consistency via Alternating Isolation**: Alternating coordinate optimization solves the bilinear aggregation inconsistency, providing strictly linear updates that enable stable, distortion-free geometric variance monitoring.
- **Decoupled Subspace Geometric Monitoring**: Evaluating drift variance directly on isolated low-rank coordinate tensors eliminates the memory and communication overhead of full-model gradient approximations.
- **Dual Frontier Operational Model**: Empirically demonstrated across 173 runs on `roberta-base`: acts as a *Compute Accelerator* in consensus (up to 56.6% bandwidth savings) and a *Safety Brake* under skew (+**26.95**% anti-collapse stability recovery on SST-2).
- **Full Empirical Foundation Benchmark**: Completed comprehensive multi-seed benchmark across GLUE tasks on dedicated Polytechnix GPU hardware.

Conclusion : Limitations & Future Research Horizons

Limitations

- **Deterministic Alternation**: Alternates A and B in strict sequence; asymmetric or adaptive switching cadences remain unexplored.
- **Simulated Multi-Process**: Executed on a dedicated multi-GPU server; physical wide-area latency and client dropout left unmodeled.
- **Backbone Scope**: Evaluated on `roberta-base` (125M); multi-billion parameter LLMs (e.g., Llama-3 8B) remain future targets.

Future Horizons

- **Advanced Controllers**: PID-regulated threshold scaling and variance-acceleration damping.
- **Broader PEFT Variants**: Formalizing FDA with FFA-LoRA, SLoRA, and dynamic rank allocation (AdaLoRA).
- **Quantization & Edge**: Synthesizing with 4-bit QLoRA for constrained edge devices.
- **Physical Edge Testbed**: Deployment across heterogeneous physical IoT and mobile hardware.

Key References

S. Chen, Y. Guo, Y. Ju, H. Dalal, Z. Zhu, and A. Khisti.
Robust Federated Finetuning of LLMs via Alternating Optimization of LoRA.
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

M. Theologitis, V. Samoladas, and A. Deligiannakis.
FDA-Opt: Communication-Efficient Federated Fine-Tuning of Language Models.
arXiv preprint arXiv:2505.04535, 2025.

E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen.
LoRA: Low-Rank Adaptation of Large Language Models.
In *International Conference on Learning Representations (ICLR)*, 2022.

B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas.
Communication-Efficient Learning of Deep Networks from Decentralized Data.
In *AISTATS*, pages 1273–1282, 2017.

I. Sharfman, A. Schuster, and D. Keren.
A Geometric Approach to Monitoring Threshold Functions over Distributed Data Streams.
In *ACM SIGMOD*, pages 301–312, 2006.

Thank You!

Questions & Discussion

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Open-Source Repository: `https://github.com/Asterinos1/FDA-LoRA`